

Storm-Gauge Forward Model (MKM-SG-001)

Wolfram Oracle Benchmark of the Code and its v2.0 Document

MKM Research Labs

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1 Executive Summary

This document benchmarks the Storm-Gauge Forward Model (MKM-SG-001): the package `src/models/stormgauge/`, which turns a storm's track, intensity and footprint into a water level at each gauge. It asks whether the code computes what its model document (`storm_gauge_v2.0.pdf`) says, and whether what the document says holds.

Version 0.1 tested the filed v1.3 of that document. The code computed its equations exactly, but 8 of the document's 16 claims failed:

- The rise was tied to the fall, so a storm never reached the levels the calibration named. It needed a peak of 56 to reach alert, and never reached severe.
- Peak level trailed peak intensity by 3.5–5 h, not 2.
- Doubling τ_{decay} lowered the peak.
- The run stopped at the storm's end, cutting durations short.
- An input was never read.
- Nothing in the platform used the model, though the document said its output fed the hazard curves.

The model was revised as v2.0:

- a rise constant of its own, 1 h;
- the recession run on;
- the unread input removed;
- the document rewritten to say what the model does and that nothing consumes it.

This version tests v2.0.

The code computes its documented equations. The haversine, the track interpolation, the kernels, the intensity at the gauge and the power law agree with a 50-digit implementation of v2.0's equations to 9.4×10^{-15}. Over 780 runs of the full response, recession included, the water level agrees to 4.4×10^{-14} m, and every peak time and duration is identical (Table 1). Wolfram, computing each independently, agrees with the reference to 4.5×10^{-31}. That includes Wolfram's own implementation of the whole response, written separately from the reference.

The document's claims hold: 16 of 16 (Table 2).

- All 169 figures in its tables reproduce from the code, including the storm response and the time constants.
- A default storm crossing the gauge reaches alert, warning and severe at the peaks it states. Peak level follows peak intensity by the 2.5 h it states. Every run ends with the gauge back near base.
- Its statements about the model are true: the speed is unread, the sphere's error is as given, the timings are as given, and nothing in the platform imports it.

Evidence. 8 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `bf0c281bd086`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-04 by `python -m tools.wolfram_oracle sg001`.

2 Model Purpose and Scope

2.1 What is benchmarked

The package v2.0 documents, imported as its own tests import it:

- `forward_model.StormGaugeModel`: the haversine, the decay kernels, the intensity at a gauge, the power law, and `compute_response`, which produces the level series and its statistics;
- `storm_factory.create_storm`, which builds the storms the cases use;
- `data_structures`: the storm, gauge and response types.

Its parameters are read from `config.models` (`_stormgauge.py`).

2.2 What is not

- Calibration of γ and the time constants to observed hydrographs, and a backtest. v2.0 states that neither exists (§4.4, §7.3), and makes them the condition of its Q9.
- `find_nearest_track_point`, which `compute_response` does not call.
- The scripted wave that writes `gaugets/`, and MKM-SS-001's sequence response.

This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements v2.0's equations directly from its text:

- the haversine of Eqs. 1–2, in its arctangent form, on a sphere of 6,371 km;
- the track interpolation of Eqs. 3–6, with α clamped to $[0, 1]$: the storm sits at its first point before the track starts, at its last after the track ends, and has no intensity after its duration T;
- the Gaussian, exponential and linear kernels of Eqs. 7–10, with $k = 1$ on the track;
- the intensity at the gauge, $I_s(t)\, k(\tilde{d}(t))$ (Eq. 11);
- the power law $\Delta L = (I/100)^{0.8} L_{\text{range}}\, s$ of Eqs. 12–13, with $L_{\text{high}} = 1.1 L_{\text{severe}}$ where a gauge has no historical high;
- the response of Eqs. 14–16 on the grid $t_i = i\,\Delta t$, $i = 0, \ldots, \lfloor (T + 5\tau_{\text{decay}})/\Delta t \rfloor$. The target is Eq. 13 at the intensity $\lfloor \tau_{\text{lag}}/\Delta t \rfloor$ steps earlier (the first step before that). It is approached at rate $\min(1, \Delta t/\tau_{\text{rise}})$ when rising and $\min(1, \Delta t/\tau_{\text{decay}})$ when falling;
- the peaks, the durations of Eq. 17 and the accumulation of Eq. 18, over the whole run.

4 Benchmark Method

The cases.

- A default storm (Table 1: 24 h, 25 track points, footprint 40 km, triangular profile, Gaussian kernel at 0.5) crosses the Thames west to east, at peaks of 30, 50, 80 and 100.
- The same storm with every other kernel and profile, and an irregular storm whose track starts late, turns and ends early.
- Five gauges: on the track, 0.1, 0.2 and 0.4° north of it, and one with a historical high and a sensitivity of 1.3.

- Four settings of $(\Delta t, \tau_{\text{lag}}, \tau_{\text{rise}}, \tau_{\text{decay}})$: one with a lag that is not a whole number of steps, and one with a step longer than the rise; and durations of 24, 23.7 and 48 h.

The closed-form families also run on distances and intensities below zero, at zero, and far beyond the footprint.

The printed tables. Tables 4, 5, 8 and 9 were generated from the code; each figure is judged as the code's value, rounded and formatted as their generator did. Tables 6 and 7 were written by hand, and are judged to the places they print.

The statements. Each statement about the model's behaviour is judged by running the code, which the checks show computes the document's equations.

- The calibration bands are solved twice: from Eq. 13 directly, and by bisection as the least peak intensity at which a default storm crossing the gauge reaches each stage.
- The recession is judged on every storm and gauge.
- An input is unread if changing it changes no response.
- Where the output goes is found by searching the tree for anything that imports the package.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the great-circle distance, as R times the angle between the two points' unit vectors (not the haversine);
- the kernels and the power law;
- the whole response, recession included, written again in the Wolfram Language for three storms. They are the default crossing, an exponential storm with a beta profile off the track, and the irregular storm. Each returns the peak level, the accumulation and the final level;
- the true distance between six UK pairs, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eqs. 1–2, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eqs. 7–10, the decay kernels: code against the reference, worst relative error	9.8×10^{-16}	1.0×10^{-12}	pass
Eqs. 3–6 and 11, the intensity at the gauge: code against the reference, worst relative error	9.4×10^{-15}	1.0×10^{-12}	pass
Eqs. 12–13, the intensity-to-level power law: code against the reference, worst relative error	8.1×10^{-16}	1.0×10^{-12}	pass
Eqs. 14–16, the response and its recession over 780 runs: worst level error (m)	4.4×10^{-14}	1.0×10^{-6}	pass
§3.7, peak level, peak intensity and accumulation (Eq. 18): worst relative error	5.7×10^{-14}	1.0×10^{-12}	pass
§3.7, peak times and durations (Eq. 17): statistics that differ	0	0	pass
Wolfram agrees with the reference	4.5×10^{-31}	1.0×10^{-25}	pass

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§3.3.1	Gaussian: 60.7% of on-track intensity at $\tilde{d} = \sigma$, 13.5% at 2σ	as printed	met
Table 4	decay by kernel and distance	21 of 21 figures reproduced	met
Table 5	the target by intensity and gauge sensitivity	15 of 15 figures reproduced	met
Table 6	$k(1.0)$ for each kernel	3 of 3 figures reproduced	met
Table 7	$(1/2)^{\gamma}$ for four exponents	4 of 4 figures reproduced	met
Table 8	a default storm by peak and gauge: peaks, lag, durations, accumulation	84 of 84 figures reproduced	met
Table 9	the time constants on the peak-100 storm	40 of 40 figures reproduced	met
§6.2	±25% in γ: +14.9% and −12.9% at $I = 50$; 10–20% over $I = 40$–60	+14.9% and -12.9%; 9.7–20.1%	met
§3.5	the target reaches alert, warning, severe at $I = 38.7$, 64.7, 86.0; the high at 100	38.7, 64.7, 86.0; 100.0	met
§3.5	a default storm crossing the gauge reaches them at peaks 40.9, 68.3, 90.7	40.9, 68.2, 90.7	met
§3.6, §6.3	peak level follows peak intensity by 2.5 h in every case of Table 8	2.5 h	met
§3.6	the run ends with the gauge back within 1% of its rise	worst 0.47% over every storm and gauge	met
§4.1	the storm's speed is not read	changing it changes no response	met
§8(5)	the sphere errs by 0.32% east–west and 0.06% north–south in the UK	-0.32% east–west, -0.06% north–south	met
Q3	0.3 ms a gauge and 8 ms for 25 (to within a factor of 3)	0.32 ms; 8.0 ms	met
§2.2, Q5, Q6	nothing in the platform imports the model	nothing in `src`, `tools` or `scripts` does	met

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented equations, and the document to describe it. Whether $\gamma = 0.8$ and the time constants fit UK catchments is an empirical question. v2.0 says so, and nothing in the repository tests it.

2. **The behavioural claims are judged on the defaults.** They are judged on Table 1's default storm and Table 2's default gauge, as v2.0 states them. A gauge with its own historical high has its own bands.

3. **Timings depend on the machine.** They are judged to within a factor of 3 of the stated figures.

7 Document History

Date	Version	Description	Author
04-Nov-2026	0.1	First issue: tests the filed v1.3; 8 of its 16 claims fail (not filed)	David K Kelly
04-Nov-2026	0.2	Tests v2.0, the revised model	David K Kelly